Cryptography & Network Security

BCSE309L

ElGamal Public Key Encryption

ElGamal Public Key Encryption

- Structure similar to RSA (discrete logs)

- Component of many secure systems
  - Digital Signature Standard (DSS)
  - S/MIME (Email)

Components of ElGamal

- Components:
  - Large prime $\boldsymbol{q}$
  - $\boldsymbol{\alpha}$ = some <u>primitive root</u> of $\boldsymbol{q}$

- $\boldsymbol{\alpha}$ is primitive root of $\boldsymbol{q}$ if
  - For all $0 < \boldsymbol{p} < \boldsymbol{q}$
    there exists some $\boldsymbol{n} < \boldsymbol{q}$ such that $\boldsymbol{\alpha}^{\boldsymbol{n}} \bmod \boldsymbol{q} = \boldsymbol{p}$
    - That is, powers of $\boldsymbol{\alpha}$ "generate" all integers mod $\boldsymbol{q}$
  - Necessary to make sure encryption has unique inverse, as this insures that $(\boldsymbol{\alpha}^{\boldsymbol{n}} \bmod \boldsymbol{q}) \neq (\boldsymbol{\alpha}^{\boldsymbol{m}} \bmod \boldsymbol{q})$ for $\boldsymbol{n} \neq \boldsymbol{m}$

Primitive Root

- Example: $q = 19$

a	a^2	a^3	a^4	a^5	a^6	a^7	a^8	a^9	a^{10}	a^{11}	a^{12}	a^{13}	a^{14}	a^{15}	a^{16}	a^{17}	a^{18}
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	4	8	16	13	7	14	9	18	17	15	11	3	6	12	5	10	1
3	9	8	5	15	7	2	6	18	16	10	11	14	4	12	17	13	1
4	16	7	9	17	11	6	5	1	4	16	7	9	17	11	6	5	1
5	6	11	17	9	7	16	4	1	5	6	11	17	9	7	16	4	1
6	17	7	4	5	11	9	16	1	6	17	7	4	5	11	9	16	1
7	11	1	7	11	1	7	11	1	7	11	1	7	11	1	7	11	1
8	7	18	11	12	1	8	7	18	11	12	1	8	7	18	11	12	1
9	5	7	6	16	11	4	17	1	9	5	7	6	16	11	4	17	1
10	5	12	6	3	11	15	17	18	9	14	7	13	16	8	4	2	1
11	7	1	11	7	1	11	7	1	11	7	1	11	7	1	11	7	1
12	11	18	7	8	1	12	11	18	7	8	1	12	11	18	7	8	1
13	17	12	4	14	11	10	16	18	6	2	7	15	5	8	9	3	1
14	6	8	17	10	7	3	4	18	5	13	11	2	9	12	16	15	1
15	16	12	9	2	11	13	5	18	4	3	7	10	17	8	6	14	1
16	9	11	5	4	7	17	6	1	16	9	11	5	4	7	17	6	1
17	4	11	16	6	7	5	9	1	17	4	11	16	6	7	5	9	1
18	1	18	1	18	1	18	1	18	1	18	1	18	1	18	1	18	1

Only primitive roots:

2

3

10

13

14

15

ElGamal Encryption Scheme

1. Key Generation protocol: Public and private keys

- **Choose a sufficiently large prime number q and the primitive root α of q.**

Example: $q = 19$, $\alpha = 10$

- Generate random integer X_A
  - $1 < X_A < q-1$

$X_A = 5$

- Compute $Y_A = \alpha^{X_A} \bmod q$

$Y_A = 3$

- Public key: $\{q, \alpha, Y_A\}$
- Private key: $\{X_A\}$

- Security similar to <u>RSA</u>
  - Must be able to solve modular logarithm $X_A = \log_\alpha Y_A$ to crack

Encrytion

- Sender creates “one time key” for encryption
  - Message encrypted using modular exponentiation on one time key and public keys
  - Key “encrypted” using modular exponentiation and public keys

  - Recipient “decrypts” one time key using their private key
  - Recipient then decrypts message using that key

2. Encryption Process:

- Sender generates random integer $k < q$
- Sender computes one-time key $K = (Y_A)^k \bmod q$
- Message M encrypted as <u>two</u> integers (C_1, C_2)

$$C_1 = \alpha^k \bmod q \qquad C_2 = KM \bmod q$$

- Example: Plaintext message $M = 17$

  Choose random $k = 6$

  $K = 3^6 \bmod 19 = 7$

  $C_1 = 10^6 \bmod 19 = \mathbf{11}$

  $C_2 = 7 \times 17 \bmod 19 = \mathbf{5}$

3. Decryption Process:

- Recipient recovers $K = C_1^{\,x_A} \bmod q$

  Example: $K = 11^5 \bmod 19 = \mathbf{7}$

- Recipient then recovers $M = (C_2 K^{-1}) \bmod q$
  - K^{-1} is <u>inverse</u> of K mod q

  Example: $7^{-1} \bmod 19 = 11 \quad (77 = 4 \times 19 + 1)$

  $$M = 5 \times 11 \bmod 19 = \mathbf{17}$$

ElGamal Encryption Scheme

Global Public Elements	
q	prime number
α	$\alpha < q$ and α a primitive root of q

Key Generation by Alice	
Select private X_A	$X_A < q - 1$
Calculate Y_A	$Y_A = \alpha^{XA} \bmod q$
Public key	$PU = \{q, \alpha, Y_A\}$
Private key	X_A

ElGamal Encryption Scheme

Encryption by Bob with Alice's Public Key	
Plaintext:	$M < q$
Select random integer k	$k < q$
Calculate K	$K = (Y_A)^k \bmod q$
Calculate C_1	$C_1 = \alpha^k \bmod q$
Calculate C_2	$C_2 = KM \bmod q$
Ciphertext:	(C_1, C_2)

Decryption by Alice with Alice's Private Key	
Ciphertext:	(C_1, C_2)
Calculate K	$K = (C_1)^{XA} \bmod q$
Plaintext:	$M = (C_2 K^{-1}) \bmod q$

Why Does This Work?

$K = (Y_A)^k \bmod q$	Definition of K
$= (\alpha^{X_A} \bmod q)^k \bmod q$	Definition of Y_A
$= (\alpha^{X_A k}) \bmod q$	Rules of modular exponentiation
$= (\alpha^{k X_A}) \bmod q$	
$= (C_1^{X_A}) \bmod q$	Definition of C_1

Why Does This Work?

$$
\begin{aligned}
\boldsymbol{M} &= (\boldsymbol{C_2}\,\boldsymbol{K}^{-1}) \bmod \boldsymbol{q} \\
&= (\boldsymbol{KM} \bmod \boldsymbol{q}\,\boldsymbol{K}^{-1}) \bmod \boldsymbol{q} && \text{Definition of } \boldsymbol{C_2} \\
&= (\boldsymbol{M}\,\boldsymbol{K}\,\boldsymbol{K}^{-1}) \bmod \boldsymbol{q} && \text{Rules of modular exponentiation} \\
&= (\boldsymbol{M}) \bmod \boldsymbol{q} \\
&= \boldsymbol{M} && \text{If } \boldsymbol{M} < \boldsymbol{q}
\end{aligned}
$$

Implementation Details

- If $M > q$ must break into smaller blocks M_1 , M_2 , M_3 … $< q$
- Must use <u>different</u> k for each block
  - Otherwise single known M_i allows <u>all</u> blocks to be decrypted by <u>factoring out</u> K
- $\frac{C_{21}}{C_{22}} = \frac{KM_1 \bmod q}{KM_2 \bmod q} = \frac{M_1 \bmod q}{M_2 \bmod q}$
- If M_1 known then $M_2 = C_{21}^{-1}\, C_{22}\, M_1 \bmod q$

Practice Problems

Problem -1

Suppose Alice and Bob use an ElGamal scheme with a common prime $q=71$ and a primitive root $\alpha=7$.

a) If Bob has public key $Y_A=3$ and Alice chose the random integer $k=2$, what is the ciphertext of $M=30$?

b) If Alice now chooses a different value of k so that the encoding of $M=30$ is $C=(59, C_2)$, what is the integer C_2?

Problem - 2: Analyzing the Security of the ElGamal Scheme

Given a prime number p=83, generator/primitive root g=2, private key x=17, and random number k=20, analyze the security of the ElGamal encryption scheme by:

a) Determining the public key.

b) Encrypting the message M=50 and finding the ciphertext.

c) Analyzing how the choice of k influences the security of the ciphertext. What happens if k is reused?